

The Arithmetic of Order: A Unified Origin for the Standard Model's Gauge Symmetries, Hypercomplex Numbers, and Codewords

Faysal El Khettabi
Ensemble AIs
faysal.el.khettabi@gmail.com

July 16, 2025

Abstract

This report details the Arithmetic of Order (AoO) as a discrete arithmetic framework that provides a unified origin for the Standard Model's gauge symmetries, hypercomplex numbers, and optimal error-correcting codes. The AoO posits that physical and mathematical structures emerge from the ordered combinatorial progression $1 \rightarrow n \rightarrow n + 1$ and its associated powerset hierarchy, $\mathcal{P}(\Omega_n)$, without assuming a pre-existing continuum. We show how this framework, realized through Farey sequences under the modular group $SL(2, \mathbb{Z})$, lifts naturally to its universal central extension, the braid group B_3 . This topological extension provides a direct mechanism where the three Reidemeister moves generate the gauge groups $U(1)$, $SU(2)$, and $SU(3)$. The discussion section contextualizes these results, highlighting the framework's power in unifying the language of physical transformations (hypercomplex numbers) with principles of information stability (codewords), and notes the convergence of our conclusions with parallel topological models of physics.

Keywords: gauge symmetry, Reidemeister moves, braid group, arithmetic foundations, hypercomplex numbers, Golay code, $SL(2, \mathbb{Z})$, Standard Model, Veldkamp Space

1 Introduction

A primary goal of fundamental physics is to understand the logical origins of the laws of nature. The Standard Model of particle physics is remarkably successful, yet it does not explain the genesis of its foundational gauge symmetries: $U(1)$, $SU(2)$, and $SU(3)$. This work presents the Arithmetic of Order (AoO) as a candidate for this explanation.

The AoO challenges the traditional reliance on continuum mathematics, which posits that finite physical systems require infinite constructs for their description. Instead, the AoO proposes that the structure of physics is an inevitable consequence of a simple, finitistic, and constructive process based on ordered distinctions. This paper outlines the results of this framework, showing how its core principles give rise to the topology of gauge interactions, and discusses the broader implications of its unifying power.

2 Core Principles of the Discrete Arithmetic Framework

2.1 Powerset Combinatorics and Finite Fields

The AoO begins with an ordered set of n distinguishable degrees of freedom, $\Omega_n = \{1, 2, \dots, n\}$. The full space of all possible system configurations is its powerset, $\mathcal{P}(\Omega_n)$, which contains 2^n unique subsets. Each subset corresponds to a specific configuration, represented empirically by a characteristic function ($\chi_S : \Omega_n \rightarrow \{0, 1\}$).

This powerset, when equipped with the symmetric difference operation (Δ), is isomorphic to the n -dimensional vector space over the finite field of two elements, $\mathbb{F}_2^n$. This provides a native algebraic language for system states based on binary logic (bitwise XOR).

2.2 Emergence of Hypercomplex Numbers and Codewords

From this single powerset engine, two critical structures emerge deterministically:

- **Hypercomplex Numbers:** The 2^n elements of $\mathcal{P}(\Omega_n)$ directly correspond to the 2^n basis units of a hypercomplex algebra. For $n = 3$, the $2^3 = 8$ elements of the powerset $\mathcal{P}(\Omega_3)$ provide a natural basis for the octonions $\{1, e_1, \dots, e_7\}$. The multiplication rules are defined by the bitwise XOR operation on the binary representations of the basis units.
- **Optimal Codewords:** When specific, finite constraints (e.g., linearity, self-duality) are applied as a "sieve" to the powerset, exceptionally stable structures emerge. The premier example at $n=24$ is the extended binary Golay code (G_{24}), a structure known for its unique error-correcting properties and its deep connection to other fundamental objects like the Leech Lattice.

In summary, the structured powerset $\mathcal{P}(\Omega_n)$ not only provides the state space of discrete systems, but also underlies the emergence of algebraic and informational tools previously seen as unrelated—namely, hypercomplex units and optimal codes.

3 Results: From Arithmetic to Gauge Theory

3.1 Arithmetic Realization: Farey Sequence and $SL(2, \mathbb{Z})$

The AoO finds a concrete realization in the Farey sequence hierarchy, $\mathcal{F}_n$. The refinement of the sequence via the mediant operation, where the mediant of $\frac{a}{b}$ and $\frac{c}{d}$ is defined as $\frac{a+c}{b+d}$, is a construction governed coherently by the modular group $SL(2, \mathbb{Z})$. This establishes a direct link between the arithmetic progression of the AoO and the algebraic symmetries of the modular group.

3.2 Topological Extension: The Braid Group B_3

The framework's key topological step relies on a well-established mathematical fact: the braid group on 3 strands, B_3 , is the universal central extension of the modular group $PSL(2, \mathbb{Z})$. This provides a rigorous, non-arbitrary pathway to lift the arithmetic process into the topological domain of braids. The braid relation, $\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2$, is precisely the third Reidemeister move.

3.3 Physical Interpretation: Reidemeister Moves and Gauge Groups

Following the interpretation advanced by Christoph Schiller, the three Reidemeister moves correspond directly to the gauge interactions. They represent the fundamental ways to deform a topological structure while preserving its essential identity. The mapping is as follows:

- **Twist (Move I)** generates the $U(1)$ group of electromagnetism.
- **Poke (Move II)** generates the $SU(2)$ group of the weak interaction.
- **Slide (Move III)** generates the $SU(3)$ group of the strong interaction.

In summary, this section establishes a formal chain of logic: the AoO's discrete arithmetic is governed by $SL(2, \mathbb{Z})$, which lifts naturally to the Braid Group B_3 , whose fundamental relations in turn map directly to the gauge groups of the Standard Model.

Proposition 1 (Farey Limit and Braid Phase Closure). *Let $\mathcal{F}_n$ denote the Farey sequence at order n , generated recursively by mediant operations governed by the modular group $SL(2, \mathbb{Z})$. Let $M_n = g_1 g_2 \dots g_k$ be a finite product of Farey refinement steps, with each $g_i \in SL(2, \mathbb{Z})$ satisfying $\det(g_i) = 1$. Then:*

$\lim_{n \rightarrow \infty} M_n$ generates a dense tessellation of the upper half-plane, $\mathbb{H}^2$, whose symmetry group is $PSL_2(\mathbb{Z})$.

The universal covering group $\overline{SL_2(\mathbb{R})}$ lifts this arithmetic tessellation to include the phase history of all braid words. The unique universal central extension of $PSL_2(\mathbb{Z})$ is the braid group B_3 , which tracks this phase memory. Hence, the braid group B_3 emerges naturally as the minimal topological phase closure of the finitistic Farey- $SL(2, \mathbb{Z})$ recursion.

Remark. In the Arithmetic of Order framework, the integers $\mathbb{Z}$ appear only as finite, ordered steps $1 \rightarrow n \rightarrow n+1$. The real line $\mathbb{R}$, and the covering group $\overline{SL}_2(\mathbb{R})$, emerge as the dense limit of the Farey sequence recursion. The braid group B_3 is the unique minimal topological extension needed to track the phase memory of this finite process. Thus, no infinite continuum is postulated: it arises as the limit of finitistic arithmetic distinctions.

Corollary 1.1. *The gauge group structure of the Standard Model, $U(1) \times SU(2) \times SU(3)$, is the necessary and minimal topological consequence of a finitistic, recursive arithmetic.*

Corollary 1.2. *The real continuum $\mathbb{R}$ that underlies relativistic spacetime emerges as the limit of finitistic, recursive distinctions in the Farey- $SL(2, \mathbb{Z})$ process. Hence, the continuum assumed in Einstein's theory of relativity is not a primitive postulate but an emergent, asymptotic consequence of the Arithmetic of Order.*

4 Discussion

4.1 A Unified Origin for Symmetries and Stability

The ability of the AoO to derive the language of physical transformations (hypercomplex numbers) and the principles of information stability (optimal codes) from the same combinatorial foundation is a powerful indicator of its potential as a unifying theory.

4.2 Robustness and Context of the Topological-Gauge Connection

The connection between the three Reidemeister moves and the $U(1)$, $SU(2)$, and $SU(3)$ gauge groups, as demonstrated by Schiller, provides a topologically robust argument for the uniqueness of the Standard Model's symmetry structure. While this framework successfully derives the *group structure*, it does not yet provide a full derivation of the Standard Model's *quantum field dynamics*. This is where the AoO provides a significant advancement: it derives the necessary Braid Group topology from more fundamental, pre-physical principles, addressing the foundational "why" behind the topological "how."

4.3 Limitations and Open Questions

While the AoO framework suggests compelling origins for gauge symmetries, it remains to be shown whether the full Lagrangian dynamics of the Standard Model—including the Higgs mechanism and fermion masses—can be derived. These questions provide fertile ground for future research.

5 Conclusion

The Arithmetic of Order demonstrates that the foundational structures of modern physics need not be arbitrary postulates. As shown, the gauge hierarchy of the Standard Model emerges as the necessary topological closure of a discrete arithmetic process. Furthermore, the very continuum of relativistic spacetime is recovered not as a given, but as the asymptotic limit of the same finitistic system. By unifying the origins of quantum gauge symmetries, relativistic geometry, and information-theoretic stability within a single, computable framework, the AoO presents a compelling and potentially revolutionary foundation for understanding the logical structure of our universe.

A Appendix: Farey Sequence Generation

The Farey sequence $\mathcal{F}_n$ is generated recursively. To obtain $\mathcal{F}_{n+1}$ from $\mathcal{F}_n$, one inserts the mediant $\frac{a+c}{b+d}$ between any two adjacent fractions $\frac{a}{b}$ and $\frac{c}{d}$ in $\mathcal{F}_n$ for which $b+d = n+1$. Example for $n=3$: $\mathcal{F}_1 = \{\frac{0}{1}, \frac{1}{1}\}$, $\mathcal{F}_2 = \{\frac{0}{1}, \frac{1}{2}, \frac{1}{1}\}$, $\mathcal{F}_3 = \{\frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1}\}$.

References

- [1] C. Schiller, "On the relation between the three Reidemeister moves and the three gauge groups," arXiv:2312.14173v1 [physics.gen-ph] (Dec 2023).

- [2] F. El Khettabi, “A Comprehensive Modern Mathematical Foundation for Hypercomplex Numbers,” (2025).
- [3] F. El Khettabi, “The Arithmetic of Order: A Finitistic Foundation for Mathematics, Emergent Structures, and Intelligent Systems,” (2025).
- [4] F. El Khettabi, “Physics as Quantized Measurement: The Farey Sequence as a Realization of the Arithmetic of Order,” (2025).
- [5] L. H. Kauffman, “Knots and Physics,” 4th ed., World Scientific, 2013.